Media Suite in the Age of AI Agents

Rethinking Access to Audiovisual Heritage

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This is a living document. Feedback via GitHub Issues:

`https://github.com/roelandordelman/mediasuite-whitepaper/issues`

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Section 1 — Introduction

For ten years, the CLARIAH Media Suite has been built on a simple but demanding premise: that researchers deserve more than a search box. They deserve structured access to audio-visual heritage — tools to search, compare, annotate, and analyse across collections, with transparency about what is available, what is missing, and how algorithmic processes shape what they see. That premise has produced something rare: a research infrastructure that is simultaneously a working platform, a technical experiment, and a community of practice.

The AI revolution has not made that premise obsolete. It has made it more urgent — and for the first time, genuinely achievable at scale.

This white paper argues that ten years of careful, standards-driven infrastructure work has put the Media Suite ahead of most institutions in its readiness for AI-assisted research access. Not because AI was anticipated, but because the decisions that make AI useful — structured data, persistent identifiers, machine-readable rights expressions, open standards for interoperability — are the same decisions that make any infrastructure good. The work was right. What changes now is the interface layer on top of it.

The shift we are describing is from a set of tools that researchers operate to a set of capabilities that an AI agent can orchestrate on a researcher's behalf. A researcher asks, in natural language: *I am studying how migration has been covered in Dutch television news over the past fifty years — where do I start, what is available, how have others approached this, and what can I access?* The answer to that question currently requires navigating multiple interfaces, consulting documentation, understanding collection structures, and knowing which tools exist. An agent built on the Media Suite's infrastructure can answer it directly — retrieving relevant collections, surfacing related research, explaining access conditions, and guiding the researcher to the right starting point — with provenance and source links at every step.

This is not a speculative vision. The components required to build such an agent are not future work — they are already under active development. An AI retrieval pipeline running on production infrastructure, an access control layer based on open standards for policy-based data access, a national dataset catalogue queryable by machine, a knowledge graph for structured reasoning over collection metadata, and concrete user scenarios in journalism and research: each of these is being built right now, in ongoing projects at the Netherlands Institute for Sound and Vision and its CLARIAH and European partners. What has been missing is the explicit framing that connects these threads — the architectural vision that shows how the pieces fit together, where the gaps are, and what the path forward looks like.

That is what this document provides.

Who this is for

This white paper is written for the Media Suite team, CLARIAH partners, and colleagues in the digital and audiovisual heritage sector who want to understand not just what we are building but why the architecture is designed the way it is — and what it implies for how similar institutions might approach the same challenge. It is not a management summary or a funding proposal, though it may inform both. It is a working document — intended to align the team, articulate the vision to partners, invite feedback from external colleagues, and be updated as the work progresses and our understanding deepens.

What this document is not

It is not an argument that the Media Suite's existing infrastructure should be replaced or fundamentally redesigned. The search infrastructure, the SPARQL endpoints, the linked data publication pipeline, the annotation and analysis tools — these remain the foundation. The AI agent layer described in this document is built on top of them, not instead of them. A decade of investment in structured, standards-compliant infrastructure turns out to be exactly the right preparation for what comes next.

It is not a claim that the transition is straightforward or that all problems are solved. There are genuine open questions — around evaluation methodology, around the practical implementation of machine-readable rights expressions, around the coordination of shared infrastructure across projects. These are acknowledged and named explicitly, because honest accounting of what remains to be done is more useful than a document that only describes what has been achieved.

It is not a finished document. The architectural thinking described here is grounded in practical prototype work — a working AI retrieval system with a knowledge graph layer and a structured evaluation framework, built to understand from the inside how the components fit together. As the team's production implementation matures and user evaluation begins, this document will be updated to reflect what we learn.

Structure

The document proceeds as follows. Section 2 briefly recounts ten years of Media Suite development — not as history but as context: what was built, what principles guided it, and why those decisions turn out to be exactly the right foundation for the next step. Section 3 describes what the AI revolution actually changes, and what it does not. Section 4 takes stock of what the Media Suite already has — the infrastructure assets that become more valuable, not less, in an agentic world. Section 5 describes the architectural vision: from pipelines to agents, from interfaces for humans to capabilities for agents, with MCP as the connective protocol. Section 6 makes the case for Open Beelden as the proving ground — the right dataset to demonstrate the vision concretely before expanding further. Section 7 shows how current project work maps to the vision. Section 8 describes a phased roadmap. Section 9 is concrete about priorities and next steps.

Section 2 — Ten years of Media Suite: the right foundation

The CLARIAH Media Suite began in 2014 with a deceptively modest ambition: to bring five prototype tools from earlier research projects into a single, sustainable environment for humanities researchers working with Dutch audiovisual heritage. Those five tools — CoMeRDa, AVResearcherXL, TROVe, DIVE, and Verteld Verleden — represented the state of the art in exploratory search, cross-media analysis, and oral history research at the time. What they lacked was a stable home: a management, development, and production environment that could outlast any single project cycle. That was what the Media Suite set out to provide.

Ten years later, having passed through CLARIAH-SEED (2014–2015), CLARIAH-Core (2015–2019), and CLARIAH-Plus (2019–2024), the Media Suite offers researchers access to more than 90 collections of Dutch cultural heritage — radio and television broadcasts, oral history interviews, newspapers, film, programme guides, audience ratings, and more — with tools for advanced search, close reading, annotation, comparison, and computational analysis. Over 400 publications, presentations, and tutorials have been produced by researchers working with or about the Media Suite. What began as an experiment has become a unique national research infrastructure with international reach.

But the significance of the ten years is not primarily the scale achieved. It is the *decisions* that were made along the way — decisions that turn out, in retrospect, to be exactly the right preparation for the moment we are now in.

Co-development as a founding principle

The most important decision came first, before any tools were built: that the Media Suite would be developed *with* researchers, not *for* them. Humanities scholars, archivists, data specialists, computer scientists, and developers worked together from the outset, guided by the recognition that mutual understanding — of technical possibilities and constraints, of archival context, of research methodology — is a prerequisite for building something sustainable. This was not just a management philosophy. It was the mechanism by which the Media Suite remained grounded in real research practice rather than drifting toward features that looked impressive but were not useful.

The practical form this took was a continuous cycle of use cases, user stories, design sessions, workshops, evaluation panels, and CLARIAH Research Fellows — researchers who conducted actual research projects using the infrastructure and reported back on what worked and what did not. Every major capability in the Media Suite today can be traced to something a researcher said they needed. The discipline of listening before building, of treating a researcher's frustration as a design requirement rather than a complaint, shaped not just the tools but the way the team thinks about infrastructure.

This co-development culture produced something harder to replicate than any individual tool: a community of practice. Researchers who understand what the infrastructure can do, who have published with it, who have taught with it, and who trust it enough to build their methodology around it. That community is itself an asset — one that an AI agent layer can serve directly, and that will be essential for evaluating whether it does so well.

The kitchen and its recipes

An early conceptual frame that proved useful was the kitchen metaphor: the Media Suite as a kitchen that provides tools (facilities) to prepare research results (meals) with workflows (recipes), data (ingredients), and documentation (cookbooks). The metaphor was useful precisely because it acknowledged that different researchers have different needs. The pizza maker who wants quick results with simple tools, the haute cuisine chef with complex workflows, and the fusion artist crossing disciplinary boundaries all needed to be served by the same infrastructure.

This led to the distinction between *close reading* — viewing, annotating, and interpreting individual items — and *distant reading* — searching for patterns across large collections, visualising distributions, analysing transcripts computationally. The Media Suite was designed to support both, and the tension between them drove many of the most important infrastructure decisions: how to build a Resource Viewer that serves a film historian examining a single clip, while also supporting a computational researcher running queries across hundreds of thousands of items via a Jupyter Notebook. Getting that balance right required constant negotiation — and getting it wrong, as happened more than once, was always quickly surfaced by researchers who expected something the system could not yet do.

The infrastructure decisions that matter now

Several decisions made during the ten years look, in hindsight, like exactly the right preparation for an agentic future — even though they were made for entirely different reasons.

Structured search with linked data. From early on, the Media Suite used linked data to enrich search results: connecting person names to the GTAA thesaurus and Wikidata, enabling disambiguation and contextual lookup within the search interface. This was done to help researchers find the right person in the right role. As a side-effect, it produced queryable structured data that an agent can traverse.

Speech transcripts with timecodes. The decision to prioritise automatic speech recognition (ASR) for radio, oral history, and news genres — and to index transcripts in Elasticsearch with timecodes — was motivated by the observation that most audiovisual material is underdescribed: an archivist's summary and a few metadata fields cannot support the kind of content-based search researchers need. The ASR pipeline, now covering large portions of the Sound & Vision archive, makes the spoken content of broadcasts findable. Those same timecoded transcripts are the raw material for temporal deep links — stable, citable references to specific moments in a programme — which no existing interface yet fully exploits.

Dataset transparency and criticism. The Collection Inspector, Dataset Archaeology, and the practice of publishing ASR quality benchmarks openly were all motivated by a commitment to research integrity: researchers should be able to understand the structure, coverage, and limitations of the data they work with. Gaps in archived genres, metadata fractures, the variable quality of speech recognition across speech types — these are documented and exposed, not hidden. This transparency is unusual among heritage institutions and it is what makes the infrastructure trustworthy enough to build AI on top of. An agent that can tell a researcher “this collection has limited metadata coverage before 1980” or “ASR quality for conversational speech is lower than for broadcast news” is providing the same kind of critical context the Collection Inspector provides — but in response to a natural language question.

Secure access with SURFconext. The decision to use SURFconext for authentication — giving Dutch university researchers single sign-on access across CLARIAH services — was the first recipe the Media Suite ever implemented, and deliberately so. Secure, auditable access to restricted material is a prerequisite for everything else. The same infrastructure now underpins the Data Access Broker and SANE (Secure ANalysis Environment) being developed for quantitative analysis of sensitive datasets. Access control is not an add-on; it is foundational.

Dataset discovery via registries. When no suitable dataset registry existed, the Media Suite built its own using CKAN. When the NDE Dataset Register emerged as the national standard for Dutch digital heritage, the Media Suite connected to it. The habit of registering datasets with structured, machine-readable descriptions — including metadata about content, coverage, rights, and access — is exactly what makes a collection discoverable by an agent rather than only by a human who knows where to look.

National and European infrastructure alignment. Throughout its development, the Media Suite was connected to SURF (for compute and authentication), DANS (for Oral History), the Humanities Cluster (for annotation infrastructure), and NDE (for dataset discovery). These connections made the Media Suite more capable than it could have been in isolation — and they created a web of interoperable infrastructure that an agent can navigate. A researcher’s question that touches on Oral History, Sound & Vision broadcasts, and newspaper collections from the KB already crosses three institutions whose data infrastructures are partially connected. The work of the last ten years is what makes that crossing possible.

Where things stood at the end of 2024 — and where they go next

At the close of CLARIAH-Plus in 2024, it was clear that the ambitions for the next phase were pointing in a consistent direction: more data science-oriented access to large datasets, safe access to sensitive data, AI applications and model training for research questions, seamless movement between qualitative and quantitative methods, personal research corpora composed from multiple datasets, and an expanded user base reaching research journalists and citizen scientists.

Every one of those aspirations was already present, and every one of them is now being addressed. The work has continued through new projects — SSHOC-NL, which is building the national data access and authentication infrastructure, and Macroscope, which is extending the Media Suite into social media and cross-media analysis — and through CLARIAH-NL as the organisation that sustains the Dutch research infrastructure for the humanities beyond any single project cycle.

What was missing in 2024 was not ambition, and not infrastructure. It was the architectural frame that names what all those aspirations have in common. That frame is the agentic architecture described in this document. The ten years of infrastructure investment is the foundation. The next phase is the interface layer that makes it accessible in the way researchers now expect — and that connects the Media Suite's capabilities to the broader AI ecosystem in a way that is principled, transparent, and true to the co-development culture that made the infrastructure worth building in the first place.

Section 3 — What the AI revolution actually changes, and what it does not

There is a version of the AI narrative that goes like this: large language models have changed everything, and institutions that do not move fast will be left behind. There is another version that goes like this: AI is mostly hype, the real work is still in the data and the infrastructure, and nothing fundamental has changed. Both versions are wrong, and both are common.

What has actually changed is specific, and understanding it precisely matters — both to avoid overreacting and to avoid underreacting. For a research environment like the Media Suite, with a decade of structured infrastructure behind it, the honest assessment is: the AI revolution changes the *interface* fundamentally, changes the *scale of what is achievable* significantly, and changes almost nothing about the *underlying requirements for good data*.

What has genuinely changed

Natural language as a first-class query interface. Before large language models, natural language interfaces to structured data were a research problem with unsatisfying solutions. Researchers had to learn query languages, controlled vocabularies, and interface conventions to get what they needed. The gap between what a researcher wanted to ask and what the system could understand was bridged by training, documentation, and frustration. Large language models close much of that gap. A researcher can now describe what they are looking for in the same language they use to write their research questions, and a well-built system can translate that into structured queries against the underlying data. This is not magic — it fails in predictable ways, and those failures need to be understood and communicated. But the direction of travel is clear and the pace is fast.

Multi-step reasoning over heterogeneous sources. The researcher question from Section 1 — *I am studying migration in Dutch television news over fifty years, where do I start?* — requires simultaneously knowing what collections exist, what their coverage is, what access conditions apply, how other researchers have approached the topic, and what tools are appropriate for the analysis. No single query to a single system answers all of that. Previously, answering it required a researcher who already knew the landscape, or hours of documentation reading. An AI agent can now retrieve from multiple sources, synthesise across them, and produce a grounded, source-linked answer in seconds. The synthesis step is what is new. Individual retrieval was always possible; orchestrated multi-source reasoning at natural language was not.

The economics of enrichment. Automatic speech recognition, named entity recognition, visual similarity, and other enrichment pipelines existed before the current AI moment. What has changed is the cost, quality, and accessibility of running them. Setting up a production-quality ASR pipeline for a large and varied archive — one that handles broadcast news, oral history interviews, and conversational speech with appropriate model choices and quality benchmarking — is still genuinely hard work. It requires years of investment in models, infrastructure, and institutional alignment, as the Media Suite's own experience demonstrates. But the barrier has dropped substantially. What this means for the agent layer is that enriched, searchable spoken content — once a long-term aspiration — is now an achievable foundation. The work of getting there deserves acknowledgment; it was not automatic.

Researcher and public expectations. This is perhaps the most consequential change of all, and the one hardest to manage. Since large language model assistants became widely accessible — from around 2022 with ChatGPT, and accelerating rapidly since — researchers, journalists, and the general public have developed strong intuitions about how AI-assisted access should feel. They arrive at a heritage research environment expecting it to work like the tools they use every day. That expectation is not always fair — heritage data is complex, rights-constrained, and cannot be treated the same way as the open web — but it is real, and ignoring it means losing users to systems that are easier to use but less trustworthy and less rich. Meeting that expectation, while maintaining the rigour and transparency that makes the Media Suite worth using, is the central design challenge of the next phase.

What has not changed

Good data requirements have not changed. An AI agent that retrieves from poorly structured, inconsistently described, rights-ambiguous data produces unreliable answers. The fundamental requirements — persistent identifiers, structured metadata, machine-readable rights expressions, documented coverage and gaps — are exactly the same as they have always been. If anything, AI makes the consequences of poor data quality more visible, because a confident-sounding wrong answer is more damaging than a transparent failure to find anything.

This is where the Digitaal Erfgoed Referentie Architectuur (DERA — roughly, a national standard for machine-readable heritage data, mandating linked data, persistent URIs, and machine-readable rights expressions across Dutch cultural institutions), becomes directly relevant. DERA defines exactly the technical foundations that make heritage data AI-accessible: persistent URIs, linked data publication using RDF and SPARQL, machine-readable rights statements via RightsStatements.org, shared terminology sources via the Termennetwerk, and a national Dataset Register for collection discovery. It explicitly anticipates machine-to-machine (M2M) interfaces as first-class citizens — not an afterthought — and its enrichment/annotation pattern maps directly onto what AI agents produce: outputs that must carry provenance metadata, be linked to objects via persistent URIs, and have their rights status declared.

The Media Suite's infrastructure decisions over the last ten years — linked data, NDE registration, structured metadata, ASR benchmarking — have been consistently aligned with DERA principles. This alignment is not coincidental and not merely compliance: it reflects the understanding that these standards exist precisely because federated, trustworthy, machine-readable access to heritage data requires them. NISV is represented on the DERA Architecture Council, which means the institution has not merely been a recipient of these standards but an active participant in shaping them — a position that carries both standing and responsibility when it comes to advocating for their consistent application. That understanding turns out to be the right preparation for an agent layer. The work of the last ten years becomes more important, not less.

The same logic extends to the European level. The common European Data Space for Cultural Heritage (DS4CH) — led by the Europeana Foundation, funded through the EU's Digital Europe programme, and now operating under a Strategy 2025–2030 — is the European counterpart to DERA: a framework for making heritage data findable, shareable, and reusable across institutional and national boundaries. NISV already contributes collections to Europeana via an aggregator. The current focus on building DERA-aligned Media Suite infrastructure offers a clear opportunity to make that contribution more structural and sustainable — connecting it directly to the same pipelines that serve researchers, rather than treating European publication as a separate obligation. The European Collaborative Cloud for Cultural Heritage (ECCCH), being built through the ECHOES consortium, complements the data space by providing the collaborative research environment where heritage data gets enriched, analysed, and transformed — the layer where AI-assisted research actually happens.

The chain runs from DERA-compliant NISV data, through the NDE Dataset Register, into the DS4CH/Europeana data space, and from there into the ECCCH research environment. An AI agent operating at the Media Suite level sits at the bottom of that chain. But what it does — making collections queryable by machine, enriching them with structured metadata and ASR, linking them via persistent identifiers — is exactly what feeds the European layer. The obligation to contribute to European data spaces is not a separate requirement sitting on top of the infrastructure work. It is the same work, viewed from a different altitude. Making the Media Suite's Europeana contribution systematic rather than occasional is both an institutional obligation and a natural consequence of building the agent layer correctly.

Data criticism is more urgent, not less. One of the most important contributions the Media Suite has made to digital humanities practice is the vocabulary and tooling for data criticism: understanding how datasets were constructed, what decisions shaped them, and how those decisions affect analysis. AI makes this more urgent, not less. A researcher using an AI assistant to explore the Sound & Vision archive needs to know that genre coverage is uneven, that ASR quality varies by speech type, that metadata fields have changed meaning over time. The agent should surface this — not bury it under a confident answer. Transparency about what the system knows and does not know, and how it knows it, is a design requirement, not an optional extra.

Access control is not a convenience feature. The complexity of rights and access in a heritage archive does not disappear because the interface is more convenient. A researcher asking an AI agent for footage from the Dutch Public Broadcaster is still subject to copyright restrictions, SURFconext authentication requirements, and institutional access agreements. The agent must respect these constraints, communicate them clearly, and route researchers to appropriate access paths — not work around them or pretend they do not exist. This is not a technical obstacle; it is a legal and ethical requirement that defines the trustworthiness of the system.

The research community still has to be in the room. The co-development culture described in Section 2 does not become less important because AI makes some things easier. If anything, it becomes more important, because the failure modes of AI systems — confident hallucination, vocabulary mismatch, uneven coverage — are less obvious to users than the failure modes of traditional search. Researchers need to be involved in defining what "working well" means, in evaluating whether it does, and in identifying the gaps and biases that automated evaluation cannot catch. User evaluation is not a phase that comes after development; it is a continuous thread through it.

The reframe: from grip on AI to AI as instrument

Managing the impact of AI — on research practice, on data governance, on institutional autonomy — is a genuine and pressing challenge. The risks are real: cost and autonomy risks from outsourcing core capabilities to large commercial AI providers, legal and ethical constraints from the AI Act and GDPR, the challenge of maintaining transparency and accountability when systems become more complex. These concerns need to be taken seriously, and the Media Suite's institutional context — as part of a public broadcaster with obligations to researchers, journalists, and the general public — makes them more pressing, not less.

But caution is not the same as passivity, and managing risk is not the same as using AI well. The position this white paper argues for is that the Media Suite — precisely because of its decade of structured infrastructure work, its commitment to transparency and data criticism, its alignment with national and European standards, and its deep entanglement with the research community — is well positioned to use AI as an instrument rather than merely manage it as a risk.

The distinction matters. Managing AI as a risk leads to defensive choices: waiting for standards to stabilise, avoiding commitments to specific models or frameworks, treating every AI application as a governance challenge first. Using AI as an instrument means making deliberate choices about where it genuinely helps — natural language access to structured heritage data is one such place — and building that capability with the same rigour and transparency that has characterised the Media Suite's development from the beginning. For an institution like NISV, whose primary asset is the archive itself, this also means being deliberate about data sovereignty: the archive is a public good, not a training dataset for commercial AI systems. Using AI as an instrument means retaining control over how archival data is accessed, processed, and enriched — keeping the institution as the authoritative source rather than ceding that role to external providers.

The infrastructure is ready. The data is largely in place. The research community exists and is engaged. What is needed now is the architectural commitment to connect these assets through an agent layer — and the evaluation discipline to know, honestly, when it is working and when it is not.

A note on what the agent layer is not

It is worth being explicit about the limits of the shift being described, because AI enthusiasm tends to obscure them.

The agent layer does not replace the tools researchers use for actual analysis. A computational researcher doing corpus analysis, a film scholar doing close reading, a data journalist running frequency analyses — these workflows are not replaced by a conversational agent. The agent helps researchers *find* the right data, *understand* what is available, and *orient themselves* within the infrastructure. The analysis itself remains the researcher's work.

The agent layer does not make access restrictions disappear. It makes them easier to navigate — by explaining what is accessible to whom, routing to the right authentication paths, and presenting options clearly. But a researcher without SURFconext access still cannot stream restricted NPO broadcasts, and an agent that implies otherwise is a liability, not a feature.

The agent layer does not eliminate the need for documentation, tutorials, and human support. It complements them. The Media Suite's investment in Help pages, How-to guides, tutorials, and data stories remains valuable — the agent draws on all of it as its knowledge base, and researchers who want to go deeper will always need the underlying documentation.

What the agent layer does is lower the threshold for researchers who are new to the infrastructure, speed up orientation for researchers who are familiar with it, and make the Media Suite's capabilities more visible and discoverable to users who do not yet know what is possible. That is a significant improvement. It is also a bounded one.

Section 4 — What we already have: the infrastructure assets

Section 3 argued that the AI revolution changes the interface fundamentally but changes almost nothing about the underlying requirements for good data. This section takes stock of what the Media Suite already has — the infrastructure assets that an agentic system needs and that years of careful work have put in place. The inventory is not exhaustive, but it is intended to be honest: each asset is described as it actually is, including where it remains incomplete or requires further investment to reach its potential.

The organising principle is simple: for each asset, what does it enable for an agent that could not be done without it?

Speech transcripts with timecodes — Elasticsearch

The Media Suite's Elasticsearch index contains ASR transcripts for large portions of the Sound & Vision archive, indexed with timecodes at the sentence and word level. This is the single most powerful asset for an agent working with audiovisual heritage, because it transforms the spoken content of broadcasts into searchable, queryable text — and crucially, text that is anchored to specific moments in time.

For an agent, this means two things. First, semantic search over spoken content: a researcher asking about the representation of a topic in Dutch television news can retrieve relevant fragments based on what was actually said, not just how the programme was catalogued. Second, and not yet fully exploited, temporal deep links: every retrieved passage has a start and end time, which means every answer the agent gives about spoken content can point the researcher to the exact moment in the broadcast rather than to the programme as a whole. This capability — stable, citable references to specific fragments of audiovisual content — is what fragment citation means in practice, and it is already technically present in the data.

Priority has been given to radio programmes, Oral History interviews, and news and current affairs genres — the content most requested for research and hardest to search without transcripts. The goal of full coverage remains ahead, but the foundation is substantial and growing.

Linked data and SPARQL — data.beeldengeluid.nl

The data platform at data.beeldengeluid.nl publishes collection metadata as linked data, accessible via a SPARQL endpoint. This is structured, queryable knowledge about the collections — not just a search index but a graph of entities and relationships: programmes, people, organisations, events, genres, time periods, and the connections between them.

For an agent, a SPARQL endpoint is a precision instrument. Where semantic search over text is good at finding relevant material when the query is fuzzy or natural-language, SPARQL is good at answering precise relational questions: which programmes feature a specific person in a specific role, across which time period, in which genre. The named SPARQL query patterns already developed in the prototype work — covering tools, collections, workflows, access conditions — demonstrate what is possible at small scale. The data.beeldengeluid.nl endpoint makes the same approach available at collection scale.

The linked data also carries rights information, connecting to the Open Beelden dataset and the broader NDE infrastructure. An agent that can query rights status via SPARQL before presenting content to a researcher is doing something qualitatively different from one that presents everything and leaves rights navigation to the user.

The NDE Dataset Register and Termennetwerk

NISV datasets are registered in the NDE Dataset Register, the national catalogue of Dutch digital heritage collections. The register is publicly queryable via a SPARQL endpoint — no authentication required — and returns DCAT-structured metadata: what datasets exist, who manages them, what their access conditions are, what distribution formats are available.

For an agent, the NDE register is the map. It answers the question "what exists and how can I reach it?" not just for NISV collections but for Dutch digital heritage more broadly. An agent tool that queries the register can tell a researcher which collections across multiple institutions are relevant to their question — before any content-level retrieval happens. This is discovery at the national level, and it is already queryable today.

The Termennetwerk — the federated search across Dutch heritage terminology sources — complements the register by providing shared vocabulary. When a researcher uses a term, the agent can resolve it against the Termennetwerk to find canonical forms, related terms, and broader/narrower concepts across the GTAA thesaurus, Wikidata, and other sources. This is the disambiguation layer that makes cross-collection search reliable rather than keyword-dependent.

SURFconext, SANE, and the Data Access Broker

The access control infrastructure — SURFconext for authentication, SRAM for role-based authorisation, SANE for trusted analysis environments, and the Data Access Broker (DAB) being developed with SURF — is the layer that makes rights-aware agentic access possible for restricted collections.

An agent that can verify a researcher's institutional affiliation via SURFconext, check their access entitlements via SRAM, and route sensitive dataset requests through the DAB to a SANE environment is doing something that most AI systems cannot: respecting the access conditions of restricted heritage data rather than working around them. This is not a constraint on what the agent can do; it is what makes it trustworthy enough to use with restricted material.

The current state of AAI is binary — SURFconext confirms that a logged-in user is affiliated with a Dutch research institution, but does not yet distinguish between institutions or grant granular access based on specific dataset agreements. SRAM adds the group and role layer that makes finer-grained authorisation possible. This infrastructure is being built and refined; it is not yet complete, but the direction is right and the investment is ongoing.

The Collection Inspector and Dataset Archaeology tooling

The Collection Inspector gives researchers — and agents — structured access to metadata about the metadata: what fields exist in a collection, what their coverage is, how completeness varies over time. Dataset Archaeology extends this to the historical decisions that shaped a collection: digitisation initiatives, changes in cataloguing policy, the introduction of thesaurus terms, metadata fractures.

For an agent, this tooling is the source of the critical context that distinguishes a trustworthy answer from a confident but misleading one. An agent that can tell a researcher "ASR coverage for this collection drops significantly before 1980" or "the genre metadata for this period reflects a cataloguing change in 2004" is providing the kind of methodological grounding that the Collection Inspector was built to support. The challenge is making this information available to the agent in a queryable form — which is an engineering task, not a conceptual one.

The research community and 400+ publications

The Zotero library of CLARIAH WP5 publications — over 400 papers, presentations, and tutorials produced by researchers working with or about the Media Suite — is a knowledge base in its own right. It documents what research questions have been asked, what methods have been used, what collections have been studied, and what findings have emerged. For a researcher starting a new project, knowing that others have worked on similar questions with similar data is enormously valuable orientation.

This corpus is already partially ingested in the prototype knowledge base developed alongside this white paper. The challenge is systematic coverage: identifying which publications describe Media Suite use, extracting the relevant methodology and dataset information, and making it retrievable by the agent. The DOI-based pipeline for publication ingestion is one of the near-term additions planned for the knowledge base.

Documentation — Help, How-to, FAQ, tutorials, data stories

The Media Suite's documentation — the Help section, How-to guides, FAQ, tutorials available via the Learn menu, and the Data Stories corpus — represents a substantial body of knowledge about how to use the infrastructure, what it can do, and what researchers have done with it. This documentation was the starting point for the prototype work described in this white paper: the first knowledge base was built entirely from the Jekyll source of the mediasuite-website repository, producing over 10,000 deduplicated chunks covering all documentation collections.

The retrieval evaluation on this knowledge base reached 86–100% Hit@10 on a structured test set of researcher questions, demonstrating that the documentation is well-suited to semantic retrieval. This is the "Ask Media Suite" capability described in the introduction — a chatbot that answers researcher questions by retrieving from the actual documentation, not from general knowledge. It is working and deployable. It is also the proof of concept for the broader architecture: the same pipeline that retrieves from documentation can retrieve from collection metadata, publication abstracts, and data stories.

Key implementation decisions from the prototype: text is chunked at approximately 1500 characters with overlap, chosen based on retrieval quality testing across chunk sizes; both semantic search and structured SPARQL queries run in parallel with results deduplicated by source URL; top-k is set to 10 with results ranked by source authority before synthesis. These parameters are published alongside the evaluation results and treated as tunable — the evaluation framework exists precisely to detect when they need adjustment as the corpus grows or query patterns shift.

Visual similarity — from prototype to production

The VisXP project, co-financed by CLICKNL, developed visual similarity capabilities for the Sound & Vision archive: shot-level embeddings that allow researchers to find visually similar content given an example image or video frame. This work proved the concept — embeddings computed, a similarity search prototype demonstrated, the technical pipeline established.

VisXP is no longer actively maintained, and the specific model it used has been superseded by more capable approaches. But the core finding stands: visual similarity search over audiovisual content is technically feasible and demonstrably valuable for heritage research. CLIP embeddings — which represent images and text in the same vector space — are the concrete implementation path forward. Available via standard model hubs, CLIP enables genuinely cross-modal queries: embedding a textual description to find visually similar content, or embedding an image to find thematically related spoken material. Several strong multilingual variants exist, making the approach viable for Dutch-language heritage collections.

What is needed is focused attention to build this capability into the agent tool set alongside the text-based Elasticsearch and SPARQL tools, so that a researcher can ask "find me footage that looks like this" and get a grounded, source-linked answer. Visual similarity is the capability that most clearly demonstrates the multimodal potential of the archive — and it is the one that most clearly distinguishes the Media Suite from text-only AI systems.

The Open Beelden dataset — with its open license and manageable scale — is the right place to build this first. It is the investment that makes the difference between a text-only demonstrator and a genuinely multimodal one.

The MCP server in development

Finally, and connecting all of the above: an MCP (Model Context Protocol) server is already under active development as part of the AI experimentation work. MCP is the open standard that allows AI agents to call tools through a consistent interface — the protocol layer that turns each of the assets above into something an agent can invoke. Tools already defined include `embed_text`, `query_archive`, and `query_milvus`. The Elasticsearch search, SPARQL queries, and vector retrieval are all being exposed as agent-callable tools through this server.

This is significant because it means the agent architecture is not being built from scratch. The connective tissue is already being put in place. What remains is the coordination work: ensuring that each tool is designed once and shared across projects rather than reimplemented per team, and that the tool catalogue grows systematically to cover each of the assets described in this section.

One constraint worth naming explicitly: the current OpenShift infrastructure operates with GPU VRAM limits (24GB) that constrain which models can run locally. The current approach — local embedding via multilingual-e5-large-instruct, with generation handled separately — works within these limits. Scaling to larger corpora or heavier generation workloads will require either hardware investment or a hybrid architecture that uses SURF HPC infrastructure for compute-intensive tasks while keeping archival data within institutional control.

Summary

None of this is hypothetical. The infrastructure exists — in varying states of completeness, but it exists. The agent layer does not need to be built from scratch; it needs to be connected, coordinated, and evaluated honestly. The gaps — visual similarity still at prototype, authorisation still binary, publication ingestion still ad hoc — are real, and they are named here because pretending they are not there would make the architecture less credible, not more. The question is not whether the foundation is right. It is. The question is whether the institutional commitment exists to finish what ten years of work has made possible.

Section 5 — The architectural vision: from pipelines to agents

The Media Suite today is a set of interfaces for humans. A researcher logs in, navigates to the search tool, constructs a query, applies filters, opens items in the Resource Viewer, saves bookmarks, creates annotations, runs a Jupyter Notebook. Each step requires the researcher to know what tool to use, how to use it, and how to interpret what comes back. The infrastructure supports the researcher; it does not yet act on their behalf.

The shift this section describes is from a set of tools that researchers operate to a set of capabilities that an agent can orchestrate — on the researcher's behalf, in response to a question asked in natural language, drawing on whichever combination of tools and data sources is relevant. The tools themselves do not change. What changes is the layer of coordination above them.

From pipelines to tools

In the current architecture, a researcher's interaction with the Media Suite follows a fixed sequence determined by the interface: search, then filter, then view, then annotate, then export. This is a pipeline: a predetermined path through a set of capabilities. Pipelines are efficient for well-understood workflows, but they break down when a researcher's question does not fit the path — when answering it requires combining results from multiple collections, consulting documentation, checking access conditions, and cross-referencing prior research, all in a sequence determined by the question rather than by the interface design.

The agentic architecture replaces the fixed pipeline with a reasoning loop. Each Media Suite capability — keyword search over Elasticsearch, SPARQL queries over the linked data endpoint, semantic retrieval from the vector index, rights status lookup, documentation retrieval — becomes a *tool* that the agent can call. The agent decides which tools to call, in what order, based on what the question requires. It can call the same tool multiple times with different parameters, evaluate whether the results are sufficient, and try a different approach if they are not.

This is not a small change. It means that the researcher's entry point is no longer a search box or a tool menu — it is a question. And the question can be as complex as the research problem requires.

MCP: the connective protocol

For the agent layer to work, each capability needs to be exposed through a consistent interface that the agent can call without knowing the internal implementation details of the underlying system. This is what the Model Context Protocol (MCP) provides.

MCP is an open protocol, developed by Anthropic and now widely adopted, that defines how AI agents connect to external tools and data sources. It works like HTTP for AI: a standard that allows any agent to talk to any tool without custom integration code. Each capability becomes an MCP server, exposing a set of named tools with defined inputs and outputs. The agent is an MCP client, calling whichever servers are registered and relevant.

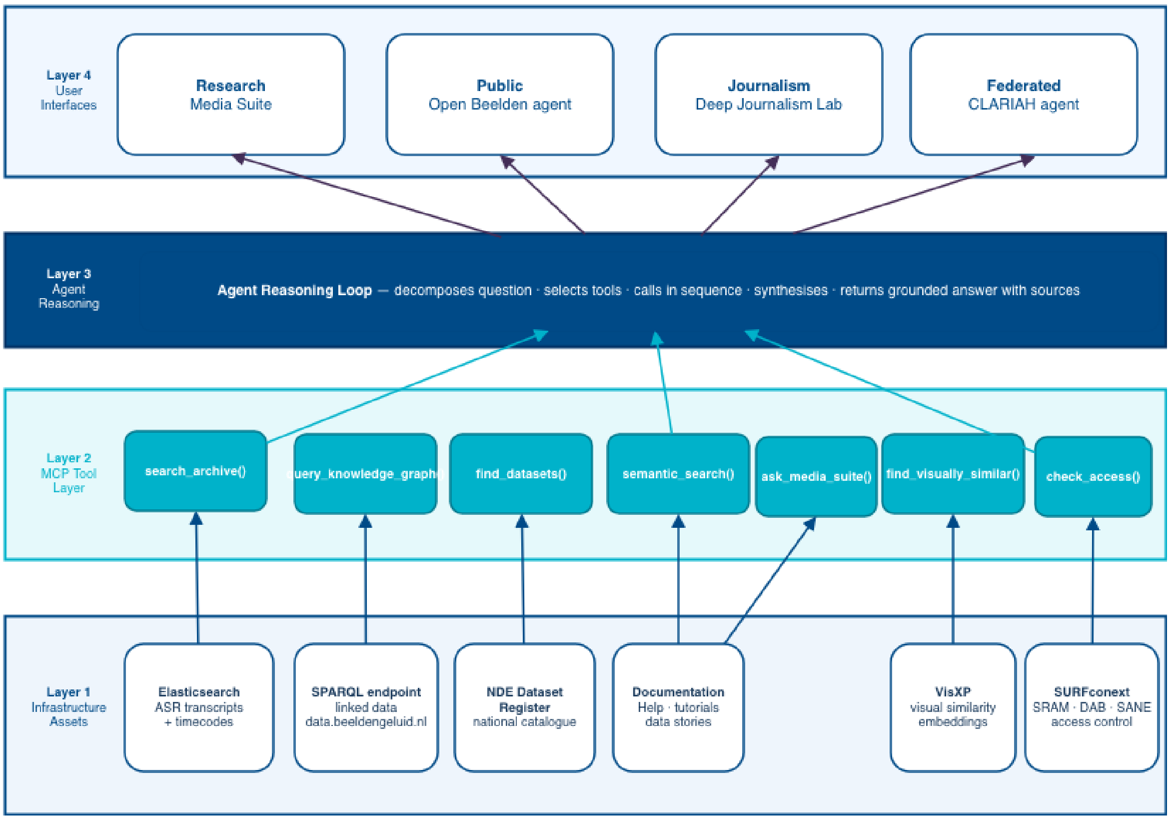

Figure 1: Figure 1: The three-layer agentic architecture. Existing infrastructure assets (bottom) are each exposed as a named MCP tool (middle layer), which an agent reasoning loop (top) calls in whatever combination a researcher's question requires. The same MCP tools serve multiple interfaces — research, public, journalistic, and federated CLARIAH.

For the Media Suite, the implication is concrete. Each capability becomes a named, callable tool:

- The Elasticsearch search: `search_archive(query, filters)`
- The SPARQL endpoint: `query_knowledge_graph(sparql_template, parameters)`
- The NDE Dataset Register: `find_datasets(institution, license, format)`
- The vector retrieval index: `semantic_search(query, collection)`
- The visual similarity index: `find_visually_similar(image_or_description)`
- The documentation knowledge base: `ask_media_suite(question)`
- The access control layer: `check_access(dataset, user_context)`

Each tool is built once and shared across any application that needs it — not just the Media Suite chatbot. A future CLARIAH tool, a researcher's own local agent, a journalistic application — all can use the same MCP servers without requiring new integration work. This is the architecture that turns a collection of project-specific implementations into shared infrastructure.

An MCP server is already under active development as part of the current AI experimentation work, with tools for embedding, archive querying, and vector retrieval already defined. The path forward is to expand this server systematically to cover the full tool catalogue described above, ensuring each tool is designed for reuse rather than for a single application.

What the agent does — a concrete scenario

The migration research question from Section 1 — *I am studying how migration has been covered in Dutch television news over the past fifty years — where do I start, what is available, how have others approached this, and what can I access?* — illustrates what the agent can do when the tool catalogue is in place.

The agent receives the question and reasons about what is needed. It queries the NDE Dataset Register to identify which collections cover Dutch television news over the relevant time period — not just NISV collections, but any registered dataset nationally. It queries the Media Suite's own collection registry to determine which of those datasets are currently indexed and queryable within the research environment. It queries the knowledge graph for structured facts: temporal coverage, genre distribution, ASR availability, access conditions. It checks the researcher's access context to determine what is immediately accessible and what requires a different route. It runs semantic search over ASR transcripts to find programmes where migration was actually discussed, not just catalogued as a subject. It queries the publications knowledge base for how other researchers have approached the question.

It synthesises all of this into a structured response: here are the relevant collections, here is what you can access now and how to request what you cannot, here is what others have found, here are the most relevant programmes to start with, and here is the documentation that will help you proceed.

Every claim is linked to its source. The researcher can follow any link, go deeper, or challenge the framing. The agent has oriented the researcher within the infrastructure — with situational awareness that previously required institutional expertise or hours of exploration — without doing the research itself.

After this orientation, the researcher typically moves into the Media Suite's familiar interface: searching within a specific collection, opening items in the Resource Viewer, building annotations, running a Jupyter Notebook. The agent does not replace this workflow. It is most useful at specific moments: at the start of a project, when the researcher does not know where to begin; at transitions, when a new tool or collection is needed; and when stuck, when a search is returning too little or too much. In between, the researcher works as before. The agent is a companion with deep institutional knowledge, not a replacement for the research process.

The registry chain — and why it has to be fixed

The scenario above makes one assumption that is not yet true and needs to be stated plainly: it assumes that querying the NDE Dataset Register gives a reliable, complete picture of what datasets exist in Dutch digital heritage, and that querying the Media Suite's local registry gives a reliable picture of what is actually available for research there. Neither assumption currently holds, and the gap between them is a structural problem that has accumulated over

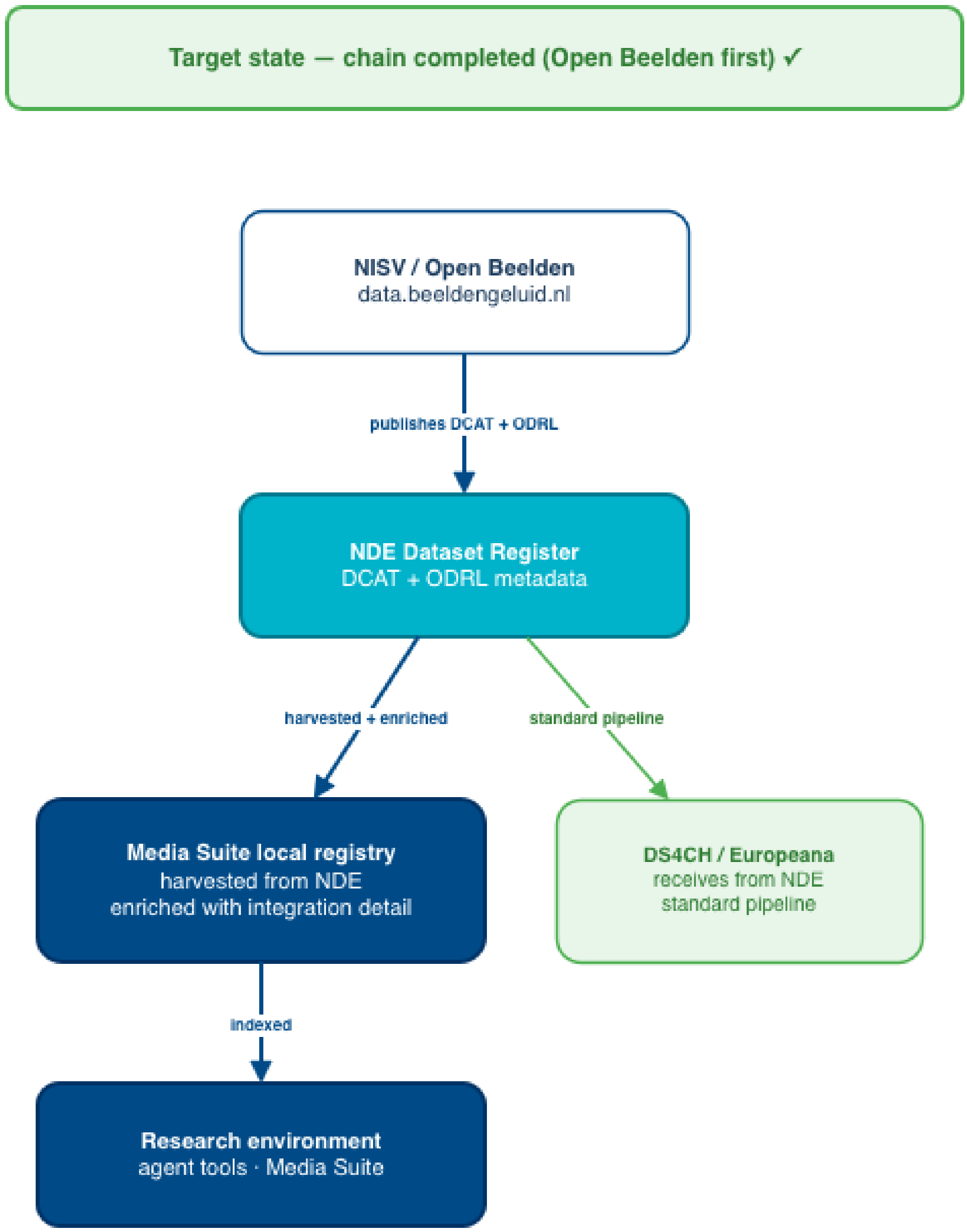

Figure 2: Figure 2: The target registry chain. NISV publishes Open Beelden to the NDE Dataset Register with correct DCAT and ODRL metadata; the Media Suite local registry harvests and enriches from NDE; the indexed data feeds the research environment agent tools; DS4CH/Europeana receives from NDE via a standard pipeline. Open Beelden is the first NISV dataset to complete this chain end to end.

years.

The intended chain runs like this: data providers register their datasets in the NDE Dataset Register with DCAT metadata and ODRL rights expressions; the Media Suite harvests from NDE to populate its local collection registry, adding integration-specific details about indexing and access; researchers and agents query both layers to find what exists and what is usable. At the European level, NDE feeds into the common European Data Space for Cultural Heritage, connecting Dutch heritage to DS4CH and making it discoverable across the continent.

The actual situation reflects a transition that has not yet completed. NISV currently contributes content to Europeana via EUscreen — the European television archives domain aggregator — a route established before NDE existed as a mature publication layer. Open Beelden content reaches Europeana through a separate route. Neither path currently goes through NDE. The practical consequence for an agent is significant: querying NDE gives an incomplete picture of what NISV holds and what is accessible, because the national layer has not yet been populated with the authoritative descriptions that should flow through it. This is not a permanent state — it is the gap that the current infrastructure work is designed to close, starting with Open Beelden.

At the national level, the Media Suite's local registry — currently implemented in CKAN — exists for a straightforward historical reason: when it was built, the NDE Dataset Register was not yet authoritative or feature-complete enough to serve as the Media Suite's primary collection catalogue. Datasets entered the CKAN registry directly, via arrangements with data providers who had neither the resources nor a strong incentive to register in NDE first. The registry accurately reflects what collections are available in the Media Suite, but it is not systematically derived from NDE — which makes it an unreliable foundation for automated discovery across the national infrastructure.

This is not a failure of technology, and it is not a failure of NDE. The standards, the register, and the network facilities have existed and been maintained. What did not happen — across the heritage sector, including NISV — was a consistent practice of treating NDE registration as the primary step rather than an optional parallel publication. The NDE register has matured considerably since the CKAN registry was established. The case for routing all new dataset registrations through it is now much stronger, and the Open Beelden integration is the point at which that practice becomes the norm rather than the exception.

The agentic architecture makes the cost of this failure more visible than it has ever been. An agent that confidently queries NDE and returns an incomplete picture is worse than no agent at all — it produces a confident wrong answer at scale. This is why the Open Beelden commitment matters beyond its immediate scope: it is the first time the chain will be completed properly, with governance enforced, for a real NISV dataset. The full chain: NISV publishes to NDE with correct DCAT and ODRL metadata; the Media Suite local registry harvests from NDE and adds integration details; the Media Suite index makes it queryable for research; DS4CH/Europeana ingests from NDE — not via a parallel aggregator route. When that chain works for Open Beelden, it becomes the model for every subsequent dataset.

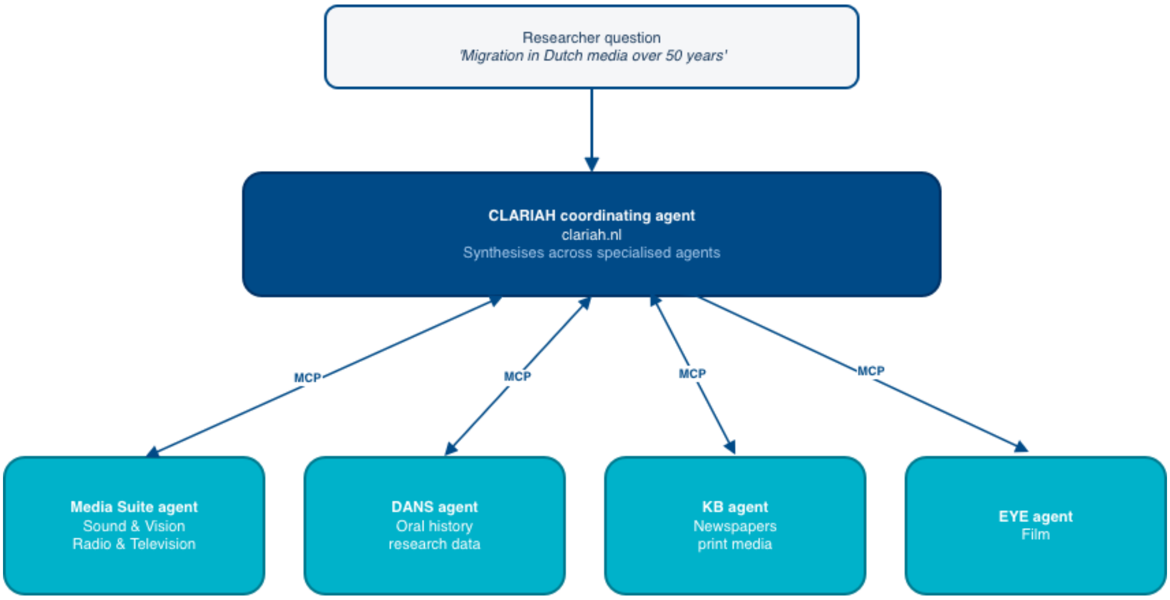

Figure 3: Figure 3: The federated agent model. Specialised dataset agents — Media Suite (Sound & Vision broadcasts and Open Beelden), DANS (oral history), KB (newspapers), EYE (film) — are coordinated by a CLARIAH-level agent that synthesises across their responses. Each specialised agent maintains its own accuracy and governance; the coordinating agent does not fabricate answers on any agent's behalf.

And when an agent queries NDE and finds Open Beelden properly described there, it will for the first time be able to give a researcher a complete and accurate answer about what that dataset contains, who can access it, and how.

Federation: the CLARIAH level and beyond

The scenario above operates at the Media Suite level — one research environment, one agent, one researcher. But the research question — *I am studying how migration has been covered in Dutch media over the past fifty years* — is not a Media Suite question. It is a question about Dutch media, which spans Sound & Vision broadcasts, KB newspapers, DANS oral history interviews, EYE film collections, and potentially social media data from multiple platforms. No single research environment holds all of this.

The right architectural response to this is federation — a network of specialised agents, each with deep knowledge of its own domain, that can be coordinated by a higher-level agent. The Media Suite agent knows the Sound & Vision collection intimately: its coverage, its access conditions, its ASR transcripts, its linked data. The DANS agent knows the oral history collections and is already developing data access infrastructure via SSHOC-NL. The KB agent knows the newspaper archive and is building towards machine-queryable access to its collections. The EYE Filmmuseum contributes film heritage. A CLARIAH-level coordinating agent — sitting at clariah.nl, or eventually replacing what INEO currently tries to be as a portal — can query all of them and synthesise across their responses. The Media Suite is not doing this alone; it is building one component of a shared architecture that several CLARIAH institutions are developing in parallel, each from their own domain.

This is not the same as building a single monolithic CLARIAH research application. That approach has been tried in various forms and consistently fails: the institutional complexity underneath does not disappear because there is a unified interface on top, and maintaining a centralised surface that accurately represents distributed, differently-governed collections requires unsustainable effort. The federated agent model is different in a crucial way: each specialised agent maintains its own accuracy and its own governance. A researcher who asks the CLARIAH agent about newspaper coverage gets an answer sourced from the KB agent's own authoritative knowledge. If the KB agent does not know something, it says so — the CLARIAH agent does not fabricate an answer on its behalf.

INEO's role in this architecture should be as a catalogue, not a portal. Its current incarnation as a unified discovery surface over heterogeneous, differently-governed collections has not delivered on its promise — the institutional complexity underneath does not disappear because there is a unified interface on top. As a machine-queryable registry of what tools, datasets, workflows, and educational materials exist across CLARIAH — the map that a coordinating agent queries to find which specialised agents exist and what they cover — it would be both more accurate and more sustainable.

The Media Suite's identity in this federated architecture is clear: it remains a research environment with its own interface, tools, and community of researchers who know it and use it directly. It also exposes its capabilities as MCP tools callable by external agents. These two roles are not in tension. A researcher who knows the Media Suite uses it as before. A researcher who asks a question on clariah.nl gets routed to Media Suite capabilities without needing to know the Media Suite exists. The institution's research environment and the federated infrastructure are the same thing, at different levels of abstraction.

What makes this achievable — rather than another promised federation that never materialises — is the combination of three things that have not previously been in place simultaneously: a standard protocol (MCP) that is simple enough for institutions to implement without large integration projects; AI agents capable of synthesising across heterogeneous sources without requiring those sources to be harmonised first; and a concrete proving ground (Open Beelden) where the full chain can be demonstrated end to end before the vision is extended. The federation does not need to be complete before it starts working. It needs one chain to work properly, and then another, and then another.

Parallel paths: vector search and structured queries

One of the key design insights from the prototype work is that vector search and structured queries are complementary, not competing. They answer different kinds of questions and should run in parallel, with the agent synthesising from whatever both return.

Vector search — embedding a question and finding semantically similar chunks of text — is good at fuzzy, open-ended questions: *what have researchers said about migration and Dutch television?* It handles vocabulary variation, finds related concepts, and works well when the researcher does not know exactly what to look for.

Structured queries — SPARQL templates run against the linked data endpoint — are good at precise, relational questions: *which programmes in the NISV archive from 1990–2010 have ASR transcripts and are accessible without a SURFconext login?* They return exact, verifiable answers and work well when the question has a definite answer.

The agent decides which path to take based on the nature of the question — or, more robustly, runs both paths and synthesises from the combination. Factual questions about collections, access conditions, and rights get routed to structured queries. Exploratory questions about content, themes, and prior research get routed to vector search. Complex questions that require both get both. The named SPARQL query catalogue — already containing eleven templates covering tools, collections, workflows, and access conditions — is the structured path's tool kit, and it grows as new question types are identified.

The dataset agent concept

The agentic architecture described above applies to the Media Suite as a whole — all collections, all tools, all documentation. But the path to building it sustainably is incremental. The concept of the *dataset agent* — a well-instrumented, independently deployable agent for a single dataset — is the building block.

A dataset agent for Open Beelden, for example, would expose: - Semantic search over ASR transcripts of the full Open Beelden corpus - Structured SPARQL queries over the Open Beelden linked data - Visual similarity search over Open Beelden keyframes using CLIP-based embeddings - Rights and access information from the NDE registration - Provenance and coverage information from Dataset Archaeology

It would be deployable independently, callable via MCP, and usable both within the Media Suite and by external applications. When a second dataset agent is built — for the Oral History collection, for the newspaper archive, for the television news collection — it is added alongside the first without changing either one. Cross-dataset queries become possible when two agents exist: *compare the framing of migration in television news and in oral history testimonies from the same period* is a question that spans two dataset agents and requires a coordinating layer to synthesise the results.

This is the dataset-by-dataset expansion model. It is deliberately incremental: prove the architecture works on one dataset, then add the next. Each addition increases the capability of the whole system without requiring a redesign. Open Beelden, as the only fully open NISV dataset with no login requirement, is the right place to start — and its migration to the new collection platform by April 2027 provides both the deadline and the motivation.

Responsible AI by design

The architectural vision described here is not neutral with respect to AI governance. Several design choices are deliberate responses to the concerns raised in Section 3.

Source grounding over generation. Every claim the agent makes is linked to a source in the knowledge base or the collection. The agent does not generate answers from general knowledge — it retrieves and synthesises from verified sources, with provenance at every step. This is not a technical constraint; it is the design choice that makes the system trust-

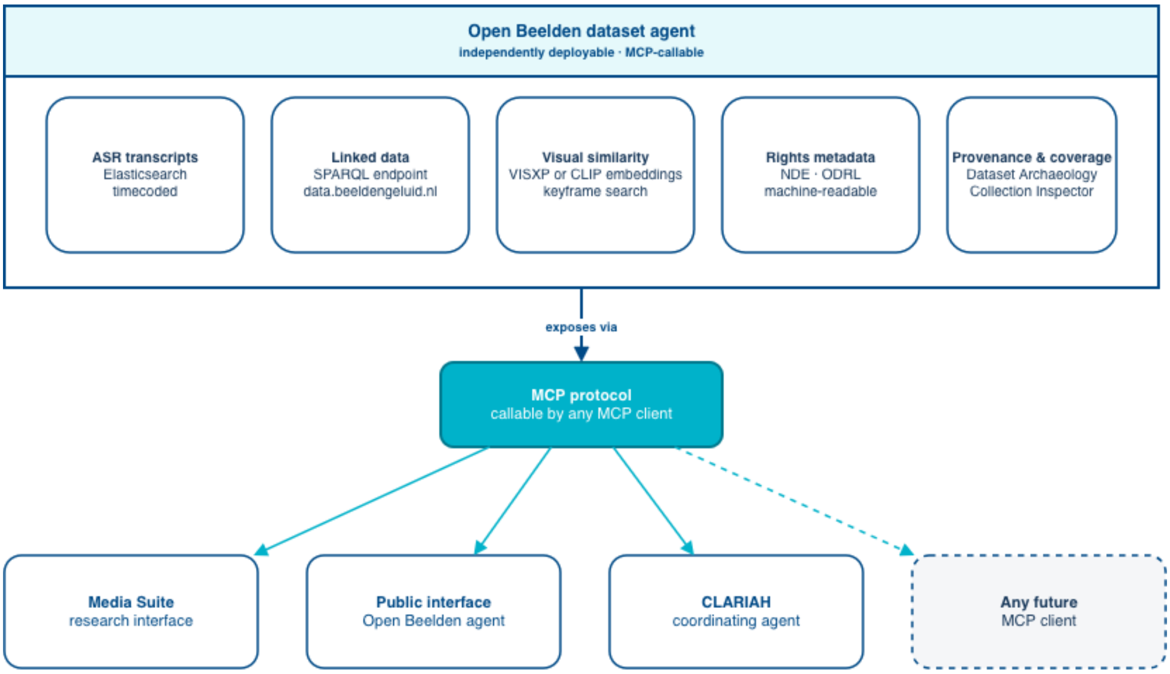

Figure 4: Figure 4: Anatomy of a dataset agent. Five instrumented components — ASR transcripts, linked data, visual similarity, rights metadata, and provenance — are bundled into a single, independently deployable, MCP-callable agent. The agent can be invoked from the Media Suite research interface, a public-facing interface, or a CLARIAH coordinating agent, without any component needing to be rebuilt per use case.

worthy for research use.

Transparency about uncertainty. When retrieval confidence is low — when the agent cannot find relevant material, or finds conflicting information, or reaches the boundary of what the knowledge base covers — it says so. Technically, this means implementing confidence thresholds: when the top-k retrieved results are semantically distant from the query, the agent falls back to an explicit acknowledgment rather than synthesising a confident answer from weak evidence. This corrective retrieval approach is built into the evaluation framework from the start — low-confidence responses are flagged, reviewed, and used to improve both the knowledge base coverage and the retrieval configuration. An answer that acknowledges its own limits is more useful to a researcher than a confident answer that conceals them.

Rights enforcement, not rights bypass. The access control layer is not a friction point to be minimised — it is a feature. The agent enforces access conditions, communicates them clearly, and routes researchers to appropriate paths. Material that requires a SURFconext login is not accessible without one, and the agent does not pretend otherwise.

Local-first where possible. The prototype work was built entirely on locally running models — no data sent to commercial APIs, no archive content used as training material. Production deployment will require infrastructure decisions about which models to run where, but the principle of keeping archival data within institutional control is not negotiable. The work to build this on NISV's own OpenShift infrastructure is already underway.

Evaluation as a continuous practice. The system is only as trustworthy as the evidence that it works correctly. The evaluation framework — structured test questions, retrieval benchmarks, publicly reported quality metrics — is built in from the start and updated as the system evolves. Quality is not asserted; it is measured and reported.

Section 6 — Open Beelden as the proving ground

Every architectural vision needs a proving ground — a bounded, real case where the theory is tested against practice, where the gaps between what is claimed and what is true become visible, and where the lessons learned shape everything that follows. For the agentic architecture described in this document, that proving ground is Open Beelden.

The choice is not arbitrary. Three advantages compound each other in a way that makes Open Beelden uniquely suited to this role, and a fourth — the hard deadline of April 2027 — makes it urgent.

Three compounding advantages

Open access — no login required. Open Beelden is the only NISV collection that is fully open: Creative Commons licensed, publicly accessible without a SURFconext login, downloadable without institutional affiliation. This matters enormously for the agent architecture because it removes the access control complexity that makes everything else harder. An AI assistant built on Open Beelden can be deployed publicly — accessible to researchers, journalists, educators, students, and the general public — without the authentication infrastructure, usage tracking, and rights enforcement that restricted collections require. This is the first opportunity to demonstrate what AI-assisted heritage access looks like when the answer is genuinely "anyone can use this," not "anyone with the right institutional affiliation can use this."

The public access angle has strategic significance beyond the technical. The Media Suite's primary audience has always been academic researchers. Open Beelden offers a route to a much broader audience — the journalist who wants to find historical footage, the educator who wants to illustrate a lesson, the citizen who is curious about Dutch media history. An agent that can answer their questions in natural language, with source links and access to the actual content, is a qualitatively different kind of public service than a traditional search interface. It is also a demonstrator for what the Media Suite could become for all open content, and what NISV's contribution to European open heritage can look like when the infrastructure is properly connected.

Already instrumented — the technical foundation is largely in place. Open Beelden is not a new dataset that needs to be built from scratch. The linked data for the B&G Open Beelden set is already present in the B&G LOD endpoint, queryable via SPARQL. ASR processing is underway for the collection, with timecoded transcripts being added to the Elasticsearch index. IIIF manifest support is being developed, which will enable fragment-level access — stable, citable temporal references to specific moments in specific videos. NDE registration exists, though it needs to be completed and enriched with proper DCAT and ODRL metadata. The collection also includes sub-collections beyond the B&G set — Natuurbeelden, VPRO, EYE, Yad Vashem — that need to be brought into the same infrastructure.

This is not a situation where the work starts from zero. It is a situation where substantial groundwork has been done and what is needed is the focused commitment to complete it

— to close the gaps, connect the layers, and bring the dataset to the standard that the agent architecture requires. The difference between "we have some linked data and some ASR transcripts" and "this dataset is fully agent-ready" is a matter of focused engineering and governance, not a fundamental rearchitecting.

NDE registration — the model for the national chain. Open Beelden is the right dataset to demonstrate the full NISV → NDE → Media Suite chain working properly, for a reason beyond its technical readiness: it is fully open, which means the governance complexity around rights and access does not obscure the technical work. A restricted dataset that goes through the chain correctly still requires careful management of who can access what and under what conditions. Open Beelden has CC licenses. The rights question is settled. This makes it possible to focus entirely on getting the technical chain right — correct DCAT metadata, correct ODRL expressions, proper distribution descriptions, NDE registration that accurately reflects what is available and how to get it — and to verify that it works end to end before applying the same approach to datasets where the rights complexity is harder.

When the Open Beelden chain is complete and working, it becomes the reference implementation. Every subsequent dataset added to the Media Suite follows the same pattern. Data providers who want their dataset in the Media Suite are shown Open Beelden as the model. NDE registration is not optional; it is the entry point. The situation with the existing CKAN registry — built before NDE was mature enough to serve as the primary catalogue — does not recur if NDE registration is the entry point from the start.

The April 2027 deadline — migration as opportunity

Open Beelden as a standalone platform (openbeelden.nl) is being phased out by April 2027. The content and functionality need to migrate to new homes: the B&G PeerTube instance for video playback, Wikimedia Commons for open distribution, and the Media Suite for research access and AI-assisted discovery.

This deadline is both a constraint and an opportunity. It is a constraint because it sets a hard date by which the Media Suite integration must be functional — the research access that openbeelden.nl currently provides cannot simply disappear. It is an opportunity because it creates institutional momentum for doing the work properly rather than quickly: the migration happens once, and if it is done with the full chain in place — NDE registration, agent-ready linked data, ASR transcripts, IIIF manifests, public access without login — it sets the standard for everything that follows.

The Media Suite team has no hard deadline responsibility for the platform migration itself. This is actually an advantage: it means there is freedom to take the time to build the agent layer correctly, rather than rushing a minimal integration to meet an operational deadline. The question is not "how do we get Open Beelden into the Media Suite by April 2027" — that is straightforward. The question is "how do we get Open Beelden into the Media Suite by April 2027 in a way that demonstrates the architecture we want for everything else."

What "done properly" means

An Open Beelden dataset agent that demonstrates the architecture needs to deliver five things, verifiably:

1. The full chain completed. NDE registration with correct DCAT and ODRL metadata, Media Suite local registry harvesting from NDE, Media Suite index built from the harvested data, DS4CH/Europeana receiving from NDE rather than via a parallel aggregator route. Not "the data is somewhere accessible" but "the chain is traceable from source to research environment to European data space."

2. Full ASR coverage with timecoded transcripts. All Open Beelden content — B&G set plus Natuurbeelden, VPRO, EYE, Yad Vashem — processed through the ASR pipeline with timecoded transcripts indexed in Elasticsearch. This is the layer that makes content-based search possible and that enables temporal deep links in agent responses. It is also the organisational coordination task that needs a named owner and a delivery date.

3. Visual similarity from prototype to production. The VisXP project established the proof of concept for visual similarity search over archival AV content, but the prototype is no longer actively maintained. The implementation path is now clear: CLIP-based embeddings, available via standard model hubs, enable cross-modal retrieval that neither requires nor depends on the original VisXP infrastructure. Open Beelden is the right dataset to build this first — manageable scale, open license, no rights complications. A researcher who can ask "find footage that looks like this" and get a grounded, source-linked answer is experiencing a capability that no other Dutch heritage interface currently offers. This is the investment that needs focused attention.

4. A public-facing agent interface. Not a researcher-only tool behind a SURFconext login but a genuinely public interface — embeddable on the Media Suite community site, accessible from a public URL, usable by anyone. The evaluation framework includes questions from journalists, educators, and general users, not just academic researchers. Quality is reported publicly, as the ASR benchmarks are, so users can understand what the system can and cannot do.

5. Evaluation as a published benchmark. A structured set of test questions — across all content types and question types that Open Beelden supports — with Hit@10 and MRR scores published openly and updated as the system improves. This is the evidence that the architecture works. It is also the discipline that prevents the "it works in demos" failure mode that has plagued previous infrastructure initiatives. If the system cannot answer a class of questions well, that is reported, not hidden.

Open Beelden as the first dataset agent

When all five of these are in place, Open Beelden becomes the first proper dataset agent in the Media Suite ecosystem: a well-instrumented, independently deployable, MCP-callable agent for a single dataset, demonstrating the full architecture from NDE chain to public-facing retrieval.

The second dataset agent — whether for the television news collection, the Oral History interviews, or a subset of the Sound & Vision archive — follows the same model. It is added alongside Open Beelden without changing Open Beelden. The cross-dataset questions become possible as soon as two agents exist: a researcher asking about migration across television news and open archival footage has two agents to query and a coordinating layer to synthesise the results.

This is the incremental path from a working prototype to a working infrastructure. Not a grand unified system, not another promised federation that will take years to materialise. One dataset, done properly. Then another. Then another.

The frustration of people who have known what was needed for years — who have watched the registry chain fail to close, who have seen linked data promise more than it delivered, who have built prototype after prototype that never reached production — is legitimate. Open Beelden is the chance to demonstrate that the frustration is over, that the architecture is right, and that the institutional commitment is real. It is a small dataset by any measure. What it proves is not.

Section 7 — The vision is already being built

The previous sections describe an architectural vision. This section makes a specific claim about it: every major component is already under active development, across multiple funded projects, by the same team. This is not a proposal for future work. It is a description of work in progress, seen from a vantage point that the individual projects do not have — because each project sees its own piece, not the whole.

For readers returning to this section directly: the core architectural components — agent layer, access control, national dataset catalogue, and knowledge graph — were introduced in Section 5 and shown in Figure 1. The table below maps these, and the cross-cutting components that support them, to the projects currently building each one. The dataset-by-dataset expansion model — one well-instrumented dataset agent at a time, each added without requiring changes to the previous — is the implementation strategy from Section 5 (Figure 4). Open Beelden, described in Section 6, is the first dataset agent and the proving ground where the registry chain (Figure 2) will first be completed end to end.

The table below maps the eight components the architecture requires to the projects currently building them.

Architecture component	Project work building it
Agent layer — embedding, vector retrieval, MCP server, agent orchestration	AI experimentation (SSHOC-NL / Macroscope)
Access control — DAB, AAI/SRAM, SANE trusted environment	National Data Flow (SSHOC-NL)
National dataset catalogue — NDE registration, DCAT, ODRL	Open Beelden Integration + National Data Flow
Knowledge graph — entity model, SPARQL query catalogue, RDF	INFINITY (Horizon Europe)
Research tool catalogue — computational methods, automatic SANE provisioning	SSHOC-NL Knowledge Graph (SSHOC-NL)
Applied user scenarios — journalism, oral history, cross-media	HAICu Deep Journalism Lab
Social media corpus — comparative cross-media analysis	Macroscope
Fragment citation — IIIF for AV, proxy fragment access	IIIF Viewer & Image Collections (SSHOC-NL)

What follows describes each component in the terms of what it contributes to the architecture, and what remains to be done.

The agent layer

The most direct implementation of the vision is already well underway. The AI experimentation work, developed jointly within SSHOC-NL and Macroscope, has produced a working pipeline: a semantic embedding layer for converting queries and collection content into comparable vector representations, a vector database deployed on NISV's OpenShift infrastructure, and an MCP server with the first agent tools already defined and callable. An orchestration layer — the component that receives a researcher's question, decides which tools to call, and synthesises a grounded answer — is in active development, with a full prototype expected by end of Q2 2026 and suitable for initial user testing.

The embedding model was selected based on multilingual retrieval benchmarks and tested against Dutch-language heritage content. It handles Dutch and English in the same vector space — important for a corpus spanning decades of Dutch-language broadcasting queried by an international research community — and outperformed alternatives on the domain-specific evaluation set used during prototype development.

The Frozen Sets concept — user-defined, topic-scoped subsets of Media Suite content that can be embedded and queried independently via RAG — is confirmed in scope. A researcher defines a dataset by specifying a topic, time period, and collection scope; the system makes it queryable as its own agent. This connects to current work in the CLARIAH community on FAIR reuse of SPARQL queries — making query definitions citable, versioned, and reusable artifacts in their own right, not just ephemeral search inputs. This points toward improved reproducibility — a Frozen Set is more than a saved search result in that it captures the corpus definition explicitly, making it easier to re-run analyses on a defined subset. Whether it fully solves the reproducibility problem is a more difficult question. True reproducibility requires that the same query returns the same results at a later date, which in turn requires that the underlying collection is versioned and that the state of the index at the time of the original analysis is preserved or citable. Collections do change: new items are added, metadata is corrected, ASR transcripts are updated. A Frozen Set as currently conceived does not address this. It is a step in the right direction — and a useful one — but the full reproducibility challenge requires versioned collections and citable data snapshots, which is a harder infrastructure problem that deserves separate attention.

The current prototype measures end-to-end query latency at 2–4 seconds for documentation retrieval on a single-user basis — acceptable for an orientation tool, though multi-user load testing under realistic conditions has not yet been conducted. Scaling to the full AV corpus will require profiling the Milvus index performance at larger vector counts and assessing whether the OpenShift GPU allocation remains sufficient or whether SURF HPC infrastructure needs to be brought in for heavier workloads. These are known unknowns, not surprises.

Two gaps are worth naming explicitly. First, a human-verified evaluation set for the agent does not yet exist. The approach that has proven effective for the documentation knowledge base — developed as a parallel prototype to ground the architectural thinking in this white paper — combines automatic validity testing with human verification: a structured set of questions is generated over the collection, automatically checked for whether retrieval returns expected sources, and then extended and validated by researchers who can judge whether the answers are actually useful. Quality metrics are reported publicly, as the ASR benchmarks are, so quality is transparent rather than asserted. This approach needs to be applied to the AV collections agent from the start, with a named owner and a commitment to publish the results. Second, the MCP server design needs to take the full architecture into account from the start — each tool built once and shared, not reimplemented per project.

The access control layer

The Data Access Broker (DAB), Authentication and Authorisation Infrastructure (AAI via SURFconext and SRAM), and SANE trusted analysis environment are being built as part of the SSHOC-NL national data flow work. These are the components that make rights-aware agentic access possible for restricted collections.

The pilot workflow — dataset registered in NDE with ODRL rights rules, user requests access via DAB, DAB verifies authorisation via SURFconext/SRAM, data transferred to SANE for quantitative research or the Media Suite for qualitative research — is a precise description of controlled agentic data access. It is being built and tested now.

Two honest caveats. Current authorisation is binary: SURFconext confirms institutional affiliation but does not yet distinguish between institutions or grant granular access based on specific dataset agreements. SRAM adds the role-based layer that makes finer-grained authorisation possible, and that work is ongoing. And ODRL implementation — machine-readable rights expressions that an agent can read to know what it is permitted to do with a dataset — is understood conceptually at NISV but not yet operationally standard. Making ODRL part of every NDE registration, starting with Open Beelden, is the concrete first step.

The national dataset catalogue

The NDE Dataset Register is already publicly queryable via SPARQL — no authentication required, no new infrastructure needed. An agent tool that queries the register can tell a researcher what datasets exist nationally, what their rights are, and how to access them. This is a working agent tool today. The NDE Termennetwerk — federated search across Dutch heritage terminology sources, including the GTAA thesaurus — is also live and queryable, providing the shared vocabulary layer that makes cross-institutional entity resolution possible. The NDE Knowledge Graph, which aggregates and connects heritage information across participating institutions, is likewise operational. These are not aspirations; they are infrastructure the agent can use now.

What is not yet working is NISV's contribution to that register. As described in Section 5 (Figure 2), NISV currently contributes to Europeana via EUscreen, bypassing NDE. The Open Beelden integration work is the vehicle for closing this gap: publishing the full Open Beelden dataset as linked data via data.beeldengeluid.nl, registering all sub-collections in the NDE register with correct DCAT and ODRL metadata, and ensuring that DS4CH/Europeana receives from NDE rather than via a parallel aggregator route.

This is the governance commitment that makes the technical architecture credible. The policy is: new datasets follow the chain. Open Beelden is first.

Fragment citation

IIIF (International Image Interoperability Framework) support for audiovisual material is being developed within SSHOC-NL's image and AV collections work. For static images, IIIF is already well established in the heritage sector — it enables stable, citable references to specific regions of a digitised object. Extending this to audiovisual material means the same principle applied to time: a stable, citable reference to a specific moment within a video or audio item, not just to the item as a whole.

For the agent architecture, this is what turns a retrieved transcript passage into a usable citation. A researcher who asks what was said about a topic gets not just a text fragment but a link to the exact moment in the broadcast where it was said — playable, shareable, and citable in a publication. Combined with the timecoded ASR transcripts already in the Elasticsearch index (described in Section 4), IIIF fragment references are the interface layer that makes the archive's spoken content genuinely accessible rather than merely findable. The IIIF work also enables proxy-based access to fragments of rights-managed content for authorised researchers, without requiring direct access to the underlying files — a meaningful step forward for restricted collections.

The knowledge graph

INFINITY is a Horizon Europe knowledge graph project for cultural heritage data, in its first year of preparation in 2026. The Media Suite's contribution is as an active partner, not a passive consumer: providing a working AV heritage test environment, contributing to the Use Case Requirements (WP5) and Ethics and Design (WP8) work packages with evidence from the prototype work, and aligning the entity model and SPARQL query catalogue with the ontology design patterns INFINITY develops for cross-institutional knowledge graphs.

The prototype knowledge graph — 1057 triples across five entity types, eleven named SPARQL query templates, deterministic embedding-based routing — is intended as an integrated test bed for INFINITY research outcomes: a real heritage collection, with real query patterns, surfacing the entity models and failure modes that production-scale knowledge graph design needs to address. The Media Suite's existing linked data assets — the GTAA thesaurus connections, the Wikidata linkages, the data.beeldengeluid.nl SPARQL endpoint — provide INFINITY with a working Dutch AV heritage environment to validate against, not just a theoretical use case. Aligning the prototype's entity model with established ontology design patterns (ODPs) — particularly for events, agents, and temporal relationships in AV heritage — is a concrete contribution the Media Suite team can make to INFINITY's shared knowledge graph design. Articulating that contribution concretely, and ensuring active participation in the relevant INFINITY work packages, is a near-term priority.

A complementary knowledge graph developed within SSHOC-NL takes a different focus: cataloguing the research tools and software methods available for computational analysis of heritage collections. Where INFINITY maps cultural heritage entities and their relationships, the SSHOC-NL Knowledge Graph maps the research landscape — which tools exist, what data types they work with, what computational environments they require, and how they connect to specific datasets.

For the agentic architecture, this has a concrete application with Open Beelden. A researcher who identifies the tools they need for a particular analysis — drawn from entries in the SSHOC-NL Knowledge Graph — can have a SANE environment automatically provisioned with exactly those tools, with the Open Beelden dataset pre-loaded by the DAB. This removes the manual dataset acquisition step from the researcher's workflow entirely: the researcher selects tools, the DAB downloads the data, the SANE environment is provisioned. The SSHOC-NL Knowledge Graph becomes part of the access control and provisioning chain — answering not just *can this researcher access this data* but *what research environment do they need to work with it*, automatically rather than by manual configuration.

The applied user scenarios

The HAICu Deep Journalism Lab — led by NISV within the HAICu programme — is developing the first end-to-end user scenario for the agentic architecture: a journalist or researcher exploring how migration has been portrayed in Dutch television, contrasted with personal testimonies from oral history collections. This scenario spans broadcast material (Sound & Vision) and oral history (in collaboration with DANS), with AI supporting cross-media retrieval, contrast, and narrative construction.

This scenario is the human evaluation that the agent layer currently lacks. Rather than depending on what the Deep Journalism Lab researchers choose to do — which may go in a journalistic rather than a research direction — the evaluation framework should be designed proactively: a curated set of questions on Open Beelden and the television news collection, structured by type and difficulty, built in the same way the documentation knowledge base evaluation was built. The Deep Journalism Lab is a valuable source of authentic questions, but it should supplement a proactively designed evaluation set rather than replace it.

The scenario also has a funding implication worth naming. The Deep Journalism Lab has no dedicated development budget for the infrastructure work the scenario requires — broadcast data pipeline, oral history integration, interactive storytelling output. A dedicated project building on CLARIAH infrastructure and incorporating Media Suite broadcast data, Oral History (in collaboration with DANS), and interactive storytelling capabilities, would be the natural vehicle. The white paper's vision provides the architectural frame for exactly that kind of project proposal.

The social media layer

The Macroscope project is building the Netherlands Media Corpus: a multi-modal, temporally structured corpus connecting broadcast (Sound & Vision), print (KB newspapers), and social media (Twi-XL and others) into a unified research resource. For the Media Suite team, the CLARIAH contribution to Macroscope is concrete: a queryable AV layer for the Netherlands Media Corpus, exposed via MCP, with an evaluation framework. This is not "AI cross-media analysis" in the abstract — it is a dataset agent that contributes the broadcast and audiovisual dimension to a federated cross-media corpus that other CLARIAH partners contribute to in parallel.

This framing also makes the legal constraints more legible. Social media data subject to platform terms of service and AVG privacy rules cannot be queried openly. The Data Access Broker and SANE trusted environment are not optional add-ons for the Macroscope corpus — they are the only compliant mechanism by which an agent can query sensitive social media data on behalf of a researcher. Macroscope is the first production test of whether DAB/SANE actually works as the agent's access control mechanism at scale.

At the CLARIAH level, the Netherlands Media Corpus is the multi-source corpus that a CLARIAH coordinating agent queries when a researcher asks a cross-media question — the kind described in Section 5's federation scenario. At the Media Suite level, the deliverable is the AV component of that corpus: queryable, MCP-callable, evaluated.

What is not yet connected

Three gaps are worth naming explicitly, because they represent coordination work that does not yet have a clear owner.

A shared evaluation framework. The agent layer, the documentation knowledge base, and the future dataset agents all need evaluation, but evaluation is being developed separately in each context. Someone needs to own the question of what "working well" means across the whole system — building the test sets, running the benchmarks, and reporting the results publicly. Concretely, this is the work of a Media Suite analyst or data scientist with domain knowledge of the collections and the research questions, working closely with researchers to extend and validate the evaluation set over time.

MCP server coordination. The agent layer is building an MCP server. The SPARQL endpoint, the Elasticsearch search, the NDE register query, the access control layer, the visual similarity index — all of these need MCP wrappers. Given a small team, the right approach is sequential and disciplined: one MCP server at a time, built to a shared standard, evaluated before moving to the next. This is not a reason for delay — the first server (the archive query tools currently in development) is already underway. It is a reason to resist the temptation to build parallel wrappers per project, and to bring in people from across the institute — on the data, legal, and rights dimensions — who need to be involved for each capability to work correctly.

The Open Beelden ASR completion. Having ASR transcripts for all Open Beelden content is technically feasible — the pipeline exists. The question is one of prioritisation and resource allocation: who coordinates running it across the full collection, including the non-B&G sub-collections (Natuurbeelden, VPRO, EYE, Yad Vashem), and by when. This needs a named owner and a delivery date before the dataset agent can be fully operational.

The summary argument

The current project portfolio is building the agentic architecture from multiple directions simultaneously. The agent layer from the AI experimentation work. The access control layer from the national data flow work. The knowledge graph from INFINITY. The first dataset agent from the Open Beelden integration. The applied user scenario from the Deep Journalism Lab. None of these projects was designed with the explicit goal of building an agentic architecture — they were designed to solve specific, funded research problems. But the architecture described in this document is what they converge on when their outputs are connected.

What is missing is not ambition and not technical capability — though the team is currently under-resourced relative to the scope of what is being built, and making that case is part of what this document is intended to support. What is missing above all is the explicit frame that names what the projects have in common and the coordination that ensures their outputs are compatible. This white paper is that frame. The coordination, and the resourcing, is what comes next.

Section 8 — Roadmap

The roadmap for this work is intended to be maintained as a living document in this paper's accompanying GitHub repository, with detailed phase descriptions, individual checklist items, and a learning log that records what changed and why. This section provides the higher-level view: the phased structure, the key decision points, and the gates that need to be passed before each phase can credibly begin.

For readers returning to this section directly: the dataset agent concept — a well-instrumented, independently deployable, MCP-callable agent for a single dataset (Figure 4) — was introduced in Section 5. The registry chain each dataset must complete (Figure 2) — NISV to NDE to Media Suite to DS4CH — was described in Section 5 and grounded in the Open Beelden case in Section 6. The three-layer architecture (Figure 1) — infrastructure assets, MCP tools, agent reasoning loop — is what each phase extends. The roadmap adds one dataset agent per phase, with gates requiring evaluation evidence before the next phase begins.

The roadmap is organised around the dataset-by-dataset expansion model: each phase adds a new dataset agent, with evaluation evidence from the previous phase informing the design of the next. This is not the only possible approach — a broader parallel build would be faster on paper — but it is the approach that is most likely to produce working infrastructure rather than another partially-completed federation.

Phase 1 — Documentation knowledge base (complete)

A working RAG pipeline over the Media Suite's documentation — Help pages, How-to guides, FAQ, tutorials, data stories — with a knowledge graph layer, dual retrieval paths (semantic search and structured SPARQL queries), and a structured evaluation framework. Hit@10 of 86–100% on a verified test set of researcher questions. Publicly deployable as an "Ask Media Suite" assistant.

This phase demonstrated the architecture at small scale and on content the team controls completely. It produced the evaluation methodology and the prototype knowledge graph that inform all subsequent phases. It is documented in the mediasuite-knowledge-base and media-suite-learn-chatbot repositories, which will be migrated to the beeldengeluid GitHub organisation as part of the Phase 4 infrastructure transition.

Phase 2 — Open Beelden dataset agent (current focus)

The full chain completed for one open dataset: NDE registration with correct DCAT and ODRL metadata, Media Suite index built from harvested linked data, ASR transcripts covering the full collection, visual similarity built using CLIP-based embeddings and deployed as a production agent tool, public-facing interface without login requirement, evaluation framework with published quality metrics.

The ambition for this phase is to have the Open Beelden dataset agent working well before any external user testing begins — the evaluation evidence from this phase is what makes subsequent phases credible.

Gate for Phase 3: the Open Beelden chain is complete, the evaluation framework shows acceptable retrieval quality, and at least one external user group (journalists, educators, or researchers) has provided structured feedback.

Phase 3 — Broadcast and television news agent

The first restricted dataset agent: a queryable agent for the television news collection, or a defined subset of the Sound & Vision broadcast archive, with access control via SURFconext/SRAM and the Data Access Broker. This phase tests the access control layer under realistic conditions — real researchers, real restricted content, real authorisation requirements.

The Deep Journalism Lab scenario — migration across television news and oral history — is both the natural evaluation context for this phase and a potential driver of which restricted dataset is prioritised first: the collection most relevant to the lab's research questions is also the one that benefits most from early agent-readiness. The Macroscope contribution — the queryable AV layer for the Netherlands Media Corpus — is the parallel development that feeds the cross-media capability.

Gate for Phase 4: the access control layer works correctly for at least one restricted dataset, the DAB/SANE pathway is tested with real researchers, and the MCP server covers both the open and restricted dataset agents with a shared interface.

Phase 4 — Prototype integration and first cross-dataset queries

The prototype repositories developed by this paper's author — built as exploratory proof-of-concept work to validate the architecture described here — are consolidated into the team's production infrastructure on NISV OpenShift and beeldengeluid GitHub. This is not a migration of the team's ongoing work, which is already being built on institutional infrastructure, but the integration of the prototype's evaluation framework, knowledge graph patterns, and MCP server designs into the shared codebase. The first cross-dataset queries then become possible: a researcher question that spans the Open Beelden agent and the television news agent, synthesised by a coordinating layer.

This phase must happen before external researcher testing at scale. It is also the phase that tests whether the institutional commitment to the chain holds when the work moves from prototype to production: registry hygiene, governance discipline, and the maintenance burden of a system that real researchers depend on.

Gate for this phase being the right moment: a natural trigger is when the first external researcher group is ready to use the system — not before, because migration under production load is harder than migration before deployment, but not after, because running production traffic on prototype infrastructure outside the institutional stack is not sustainable.

Phase 5 — Knowledge graph at scale and CLARIAH federation

The INFINITY project provides the production-scale content knowledge graph. The NDE register, properly populated by phases 2 and 3, is the catalogue a CLARIAH-level coordinating agent can query. The Media Suite agent becomes one component in a federated network — the first time the architecture described in Section 5 is actually tested at scale rather than demonstrated in a prototype.

The SSHOC-NL Knowledge Graph adds the tool layer: researchers browsing available computational methods, with automatic SANE provisioning and DAB-mediated dataset download as the delivery mechanism. By this phase, the Open Beelden precedent — tool selection from the SSHOC-NL KG triggering automatic environment setup — can be extended to additional NISV collections as they enter the NDE chain.

Whether a CLARIAH coordinating agent actually exists by this point, or whether it is being built in parallel, is an open question. What is not open is that the Media Suite's MCP tools need to be ready for it — callable by any agent, not just the one we build ourselves.

Phase 6 — Real users, real feedback

Structured evaluation with researchers who were not involved in building the system. This is where the architecture either proves itself or reveals its assumptions. The evaluation framework built in phases 1 and 2 is the scaffold; the researchers are the test. Published results, updated as the system improves, honest about failure modes. The open question here is not technical — it is whether the evaluation discipline is maintained when the results are uncomfortable.

Phase 7 — The model becomes available

When the dataset agent approach is working for two or three NISV collections, other institutions will want to use the same model. This is not a separate project; it is the consequence of doing the earlier work well and documenting it honestly. The NDE chain, the MCP server design, the evaluation framework — these become a reference implementation. Whether CLARIAH or NDE takes ownership of maintaining and promoting that reference is a governance question that phase 7 will force into the open.

Key dependencies and risks

The governance commitment. Every phase depends on the NDE chain being maintained and extended. If the governance discipline established for Open Beelden in phase 2 is not sustained — if subsequent datasets bypass the chain, if NDE registration slips back to being optional — the federated architecture fails regardless of the technical quality of the individual components. This is the risk that has caused previous federation attempts to fail, and it is the risk that requires active management, not just technical implementation.

The resourcing gap. The team is currently under-resourced relative to the scope of what is being built across multiple funded projects simultaneously. The evaluation framework, the MCP server coordination, the visual similarity build, the Open Beelden ASR completion — each of these needs sustained attention from someone with the right combination of technical and domain knowledge. The white paper is partly intended to make this case: the architectural vision is credible and the project portfolio is aligned, but the coordination and evaluation work requires investment that is not fully covered by current project budgets.

Prototype integration timing. Consolidating the prototype work into the team's production codebase (Phase 4) should happen as soon as the Open Beelden evaluation evidence is strong enough to guide the design choices — not before, to avoid integrating something that needs further iteration, and not long after, to avoid the prototype diverging from the production direction.

The visual similarity gap. The VisXP project established visual similarity as technically feasible for archival AV content, but the prototype is no longer maintained. The decision to build CLIP-based visual similarity into the Open Beelden agent — using current models available via standard model hubs — needs to be made early in phase 2, not deferred to phase 3. If the Open Beelden dataset agent launches without visual similarity, it demonstrates a diminished version of the architecture's potential.

This roadmap is a living document

The roadmap described here, and maintained in detail in `ROADMAP.md`, will change as the work progresses. What we learn from the Open Beelden evaluation will change how we approach the television news agent. What we learn from the Deep Journalism Lab will change how we design the evaluation framework. What we learn from the NISV migration will change what we recommend to other institutions.

This is the epistemically honest position: the architecture is right at the level described in this document, but the implementation details will be revised in the light of evidence. The learning log in `ROADMAP.md` records what changed and why. The version history of this document records how the vision itself evolves.

The goal is not to produce a document that is correct once and then maintained forever unchanged. It is to produce a document that is honest about what is known, clear about what is uncertain, and updated as the uncertainty resolves.

Section 9 — Priorities, next steps, and what this enables

For readers returning to this section directly: the three priorities below are direct consequences of the architectural commitments made earlier in this document. Priority 1 — completing the Open Beelden chain — delivers the dataset agent proving ground described in Section 6, including the full registry chain (Figure 2) and the five-component dataset agent structure (Figure 4). Priority 2 — coordinating the MCP server — builds the connective tool layer shown in Figure 1 (Section 5). Priority 3 — the evaluation framework — makes the responsible AI design principles from Section 5 operational rather than aspirational. Section 8 sets out the phased roadmap within which these priorities sit.

The three things that matter most right now

Across the sections of this document, several priorities emerge. Not everything can happen at once, and the team is small. These three are the ones where decisions made now have the most consequence for everything that follows.

1. Complete the Open Beelden chain — properly. This is the work that makes everything else credible. NDE registration with correct DCAT and ODRL metadata. ASR transcripts across the full collection, with a named owner and a delivery date. Visual similarity built into the Open Beelden agent using CLIP-based embeddings — a decision that needs to be made now, not deferred. A public-facing interface with a published evaluation framework. This is not one project among many; it is the proof of concept that the entire architecture rests on.

2. Coordinate the MCP server — as shared infrastructure. The mediasuite-agent MCP server is being built. The SPARQL endpoint, the Elasticsearch search, the NDE register query, the access control layer, and the visual similarity index all need MCP wrappers. The discipline required is sequential and deliberate: one capability at a time, built to a shared standard, evaluated before the next. The coordination across projects and teams — ensuring each wrapper is built once and shared rather than reimplemented — needs active ownership. This is an architectural governance task as much as a technical one.

3. Build the evaluation framework as a public benchmark. Every capability added to the system needs a test set, a quality metric, and a published result. Not after deployment — from the start. The model is the ASR benchmarking at opensource-spraakherkenning-nl.github.io: transparent, updated as the system improves, honest about failure modes as well as strengths. The evaluation framework built for the documentation knowledge base prototype is the starting point. Extending it to cover Open Beelden content, then restricted collections, then cross-dataset queries, requires sustained attention from someone with both domain knowledge and evaluation methodology skills. This role needs to be filled.

What the team needs to do

These priorities require trade-offs. Not everything currently in the backlog can proceed at the same pace. The evaluation framework and MCP coordination are coordination investments that pay dividends across all projects — which is precisely why they justify deprioritising some project-specific feature work to make room. Naming that trade-off explicitly is part of what makes the priorities credible.

Immediately: - Confirm a named owner for the Open Beelden ASR completion, with a delivery date, covering all sub-collections including Natuurbeelden, VPRO, EYE, and Yad Vashem - Make the visual similarity decision: implement CLIP-based visual similarity for Open Beelden, or explicitly defer with a stated reason - Agree on the MCP server design standard before additional wrappers are built

In the next quarter: - Complete the Open Beelden NDE registration with correct DCAT and ODRL metadata - Launch the public-facing Open Beelden agent interface, even in beta, with the evaluation framework running alongside it - Establish the evaluation framework as a shared responsibility — not attached to any single project — with results reported publicly

Over the next year: - Use the Open Beelden evaluation evidence to design the television news agent, with the Deep Journalism Lab scenario informing which collection is prioritised - Integrate the prototype repositories into the team's production codebase once the design choices from Open Beelden are stable - Articulate NISV's concrete contribution to INFINITY ahead of the next project meeting — using the prototype knowledge graph as evidence

What this enables beyond the research community

The focus of this document has been the research use case: the historian, the computational humanities researcher, the digital journalist who needs to navigate a complex heritage archive. That is the right starting point — it is the audience the Media Suite was built for and the one that has the clearest requirements and the most experience with the infrastructure.

But the architecture described here is not specific to research. The MCP servers, the agent tools, the NDE chain, the evaluation framework — these are general-purpose infrastructure. What changes between audience types is not the infrastructure but the interface: the system prompt, the tool priorities, the response style, the level of provenance detail required.

Broadcast professionals. A producer or archivist looking for reusable historical footage needs fast, reliable rights clearance and high-quality preview — not a methodological discussion of source criticism. The same agent tools that tell a researcher "this collection is accessible under SURFconext with these restrictions" can tell a broadcast professional "this footage is available for reuse under this license, here is the contact for licensing." The archive query tool is the same; the response framing is different. As NISV builds AI-assisted services for the broadcast market, the Media Suite's agent infrastructure is the natural foundation — not a separate system to build from scratch.

Educators. A teacher building a lesson on Dutch media history needs age-appropriate framing, curriculum alignment, and confidence that the content is accessible without institutional login. Open Beelden, with its CC licensing and public access, is already the right dataset. An educator-facing agent interface — a different system prompt, a different response format, the same underlying tools — is a modest extension of what the research-facing interface builds. The "Data als Kans" workshops and the Media Suite's existing tutorial infrastructure are the natural community for a pilot.

The general public. The citizen who is curious about Dutch media history, or who wants to find historical footage of an event they remember, or who is exploring their family's cultural background through archival material — this audience has been the stated aspiration of Open Beelden since its founding. An AI assistant that can answer "what footage exists of the 1953 flooding" or "show me how Dutch television covered the moon landing" is a qualitatively different kind of public access than a search box. It is also, for the first time, genuinely achievable without a specialist interface or institutional expertise.

The important point is that building these audience-specific interfaces does not require rebuilding the underlying infrastructure for each one. The MCP servers, the evaluation framework, the NDE chain — these are built once, in the research context where the requirements are best understood, and then reused across audiences. This is the argument for investing in the infrastructure properly: not because research is the only audience that matters, but because research is the right starting point for infrastructure that ultimately serves everyone.

A final note on the prototype

The architectural thinking in this document is grounded in practical prototype work: a working AI retrieval system over the Media Suite's documentation, with a knowledge graph layer, dual retrieval paths, a structured evaluation framework, and a chat interface deployable on the community site. It was built by this paper's author as exploratory proof-of-concept work — undertaken to connect the dots through direct experience rather than through reading about it, and separate from any specific funded project deliverable. The two repositories involved are publicly accessible and will be integrated into the team's institutional infrastructure as part of Phase 4.

It produced lessons that could not have been anticipated from the architecture documents alone. Retrieval quality is only as good as the test set used to measure it, and the test set is only as good as the domain knowledge used to construct it. The governance failures that have plagued linked data federation are real, not theoretical, and naming them honestly is the prerequisite for doing better. The MCP protocol adds a tool-call layer above existing standards — SPARQL, IIIF, REST APIs — that makes agent-mediated access tractable: not by replacing those standards, but by giving AI agents a consistent, discoverable interface to invoke them. And the gap between a working prototype and production infrastructure is primarily a governance and resourcing gap, not a technical one.

These lessons are in the white paper because they are the most useful things it can offer to people who will build the next stages. The architecture is right. The path is clear. The work is already underway. What remains is the institutional commitment to see it through — dataset by dataset, evaluation by evaluation, chain link by chain link — until the federation that the heritage sector has been promising itself for twenty years finally materialises, one working system at a time.

Revision history

Version	Date	Changes
0.1.5	May 2026	Sections 7, 8, 9: precision and consistency revision — orientation reminder updated; prototype/personal-project language clarified for external readers throughout Phases 1, 4, and final note; ROADMAP.md reference softened; MCP/federation claim narrowed; "What the team needs to do" structure fixed. Issue #6.
0.1.4	May 2026	Visual similarity: VisXP framing updated throughout to acknowledge prototype is superseded, CLIP-based embeddings positioned as path forward. SSHOC-NL Knowledge Graph added to Section 7 and Phase 5. Issues #1, #2.
0.1.3	May 2026	Section 7: removed internal filenames and tool identifiers; fixed table column header; added fragment citation subsection. Issue #4.
0.1.2	May 2026	Rephrased research question (three occurrences); reframed CKAN registry history in Sections 5 and 6. Issues #3, #5.

Version	Date	Changes
0.1.1	May 2026	Added four architectural diagrams (Figures 1–4); added orientation reminders at the opening of Sections 7, 8, and 9. Issue #7.
0.1	May 2026	Initial draft. Sections 1–9 complete.